



# Project Overview & System Architecture

## MULTI-PLATFORM BIOMETRIC WORKFORCE & ATTENDANCE ECOSYSTEM

A comprehensive technical blueprint and architectural specification of TimeLogic — encompassing DeepFace computer vision, zero-downtime offline kiosks, cryptographic dynamic QR tokens, automated penalty engines, and enterprise cloud infrastructure.

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### ARCHITECTURE

Distributed Edge + Cloud  
Microservices

### COMPUTER VISION

DeepFace 512-d  
Biometric AI Engine

### OFFLINE CORE

IndexedDB / SQLite  
Autonomous Outbox

### SECURITY

30-day JWT + HMAC  
Dynamic QR Tokens

# A unified workforce platform engineered to eradicate proxy attendance and payroll fraud.

TimeLogic bridges the gap between physical workplace presence and digital payroll automation. By unifying specialized edge hardware terminals with a high-availability cloud backend, TimeLogic delivers instantaneous, tamper-proof workforce tracking across five interconnected applications:

## KIOSK STATION PWA 2.0

### Contactless Facial Kiosk

React 18 / Vite PWA running on wall-mounted tablets and all-in-one PCs. Features 800ms biometric matching, privacy search mode, audio pings, and complete offline outbox autonomy.

## DESKTOP ADMIN APP

### Station & Supervisor Console

Electron-powered cross-platform desktop application (Windows .exe, Arch Linux/Pacman, Debian .deb). Enables on-site supervisor manual overrides, student tracking, and live audit wallboards.

## MOBILE EMPLOYEE APP

### Hybrid & Field Worker Client

React Native / Expo Android application. Features dynamic 30-second rotating QR scanning, GPS geofencing perimeter validation, shift requests, and personal attendance ledgers.

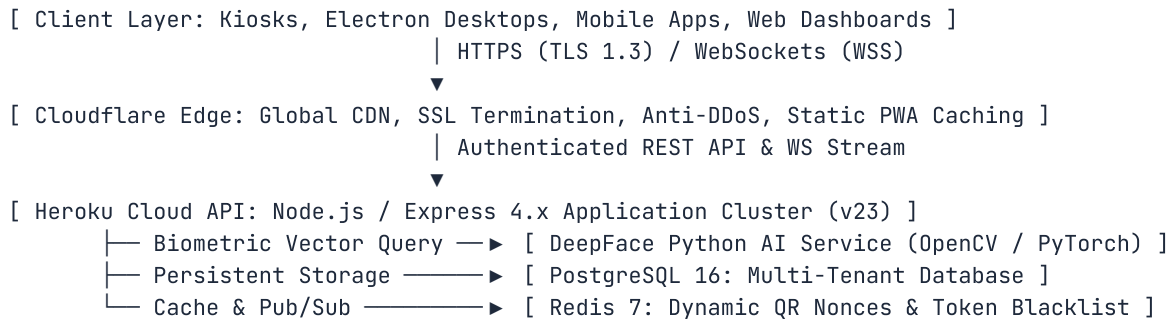
## SUPER ADMIN & WEB PORTAL

### Enterprise Control Center

Next.js / React management portal for company HR directors and multi-tenant platform administrators. Governs shift policies, late penalty tiers, biometric enrollment, and payroll sync.

**Core Technical Differentiator:** While legacy attendance systems fail immediately during power fluctuations or ISP drops, TimeLogic guarantees continuous shift verification and local headcount integrity through its zero-downtime offline architecture.

Distributed edge-to-cloud architecture designed for high concurrency and sub-50ms live updates.



### Real-Time Event Propagation via WebSocket

Every confirmed punch, manual override, or security flag is immediately broadcast through a centralized WebSocket hub in <50ms, updating supervisor live wallboards across branches instantaneously without polling overhead.

### Multi-Tenant Data Isolation

All database queries strictly enforce organization-level tenant partitioning (`company\_id` scoping), preventing data leakage between corporate clients across shared cloud infrastructure.

### 03 — COMPUTER VISION & DEEPFACE INFERENCE

## High-speed facial verification driven by 512-dimensional embedding vectors and live anti-spoofing.

TimeLogic incorporates an advanced biometric microservice utilizing the state-of-the-art DeepFace framework (Facenet512 / VGG-Face / ArcFace backends) to perform identity verification without storing raw unencrypted facial images.

#### VECTOR EMBEDDINGS

##### 512-d Mathematical Signatures

During onboarding, 3 reference facial frames are transformed into a normalized 512-float vector. Only irreversible numerical representations are stored in PostgreSQL, ensuring total privacy.

#### ANTI-SPOOF LIVENESS

##### Passive Texture & Depth Analysis

Neural network heuristics detect micro-motion, spectral frequency distributions, and moiré screen patterns, immediately rejecting printed photos, tablet screens, and silicone masks.

#### VECTOR SEARCH

##### Cosine Similarity Scoring

Vector distance is evaluated via cosine similarity:  $\text{sim} = (A \cdot B) / (||A|| \cdot ||B||)$ . High-security gates enforce strict thresholds ( $>0.65$ ), while standard kiosks operate at 0.60.

#### EDGE OPTIMIZATION

##### Sub-800ms Latency Budget

Client-side lightweight Haar/SSD cascades detect face bounding boxes before frame transmission, slashing payload size and achieving total matching throughput under 800 milliseconds.

## Autonomous local persistence guarantees attendance tracking during total network blackouts.

A critical architectural pillar of TimeLogic is its absolute resilience against Internet Service Provider drops, fiber line cuts, and local power instability. When the station terminal loses connectivity, it operates completely autonomously:

### **IndexedDB & SQLite Local Outbox Storage**

All check-in transactions occurring during an outage are cryptographically timestamped and queued locally in the browser's persistent IndexedDB or Electron's SQLite database under the `attendance\_outbox` table.

### **Optimistic Real-Time Headcount Integrity**

The kiosk UI maintains optimistic local headcount counters (e.g. 7/7 present) by overlaying queued outbox punches on top of the cached employee roster. Workers and floor supervisors receive immediate, verified confirmation without delay.

### **Idempotent Batch Synchronization Protocol**

A background reconnection daemon polls the server every 15 seconds. Upon network restoration, all queued punches are dispatched to `POST /api/station/attendance/batch-sync`. The backend verifies client UUIDs, prevents duplicate punches, updates PostgreSQL, and clears the outbox.

### **Extended Token Longevity & Auto-Rotation**

Station terminals are issued 30-day Station JWT access tokens backed by 90-day Refresh Tokens, ensuring kiosks remain operational and authenticated even through extended multi-week site disconnects.

## Cryptographic mobile clock-in combined with automated tardiness and payroll deduction logic.

### DYNAMIC ROTATING QR

#### 30-Second HMAC TOTP Tokens

The kiosk screen renders a rotating cryptographic QR code regenerating every 30 seconds. Encrypted payloads contain station UUID, UNIX epoch, and single-use nonces to defeat photo/video relay attacks.

### GEOFENCE VALIDATION

#### Haversine GPS & Wi-Fi BSSID

When a mobile worker scans the code, device GPS coordinates are validated against the branch boundary (<50m radius) alongside authenticated enterprise Wi-Fi router BSSIDs.

### RULES ENGINE

#### Multi-Tier Penalty Deductions

Arrivals past the shift grace period trigger automated tiered deductions: Tier 1 (warning), Tier 2 (30-min pay deduction), Tier 3 (1-hour pay deduction), calculated directly against hourly pay rates.

### PAYROLL STUDIO

#### One-Click ERP Synchronization

Pre-built connectors synchronize finalized hours, overtime multipliers, and late penalties into Gusto, QuickBooks, Paystack, and ADP, eliminating hundreds of manual HR calculation hours.

**Dispute & Waiver Workflow:** Employees can submit digital late waivers with documentation directly from the mobile app. HR approval instantly reverses financial penalties in the immutable payroll ledger.

## Centralized terminal enrollment, remote seat governance, and real-time fraud telemetry.

### **Cryptographic Hardware Device Binding**

Kiosk machines are enrolled using single-use Station Setup Keys. The backend extracts hardware fingerprints (WebCrypto platform UUIDs, screen architecture) to bind the physical device to the tenant. Devices cannot be cloned or moved without administrative re-authorization.

### **Instant Remote Station Revocation**

If a tablet or kiosk is lost, damaged, or decommissioned, an administrator taps "Revoke Device" in the web dashboard. The station ID is immediately pushed to the Redis distributed token blacklist, severing session access across the cluster in <1 second.

### **Tamper-Evident Append-Only Audit Trail**

Every administrative action (manual punches, shift modifications, waiver approvals, threshold adjustments) is written to an immutable audit ledger containing operator ID, ISO timestamp, IP address, and previous/new state values for full compliance reviews.

### **Automated Fraud Heuristics**

The security engine continuously evaluates real-time punch feeds, automatically flagging impossible travel velocities (punches at distant branches within minutes), repeated failed biometric liveness attempts, and duplicate punch bursts.

## 07 — PRODUCTION INFRASTRUCTURE & SLA GOVERNANCE

# Enterprise cloud infrastructure engineered for 99.99% operational continuity.

### CLOUD INFRASTRUCTURE STACK

#### High-Availability Container Topology

Deployed across redundant Heroku Eco-Dynos and managed PostgreSQL clusters (v23 release). Traffic is routed through Cloudflare Enterprise CDN with automatic edge SSL termination, Brotli compression, and DDoS protection.

### CONTINUOUS DELIVERY PIPELINE

#### Zero-Downtime Rolling Releases

GitHub Actions CI/CD pipeline triggers automated unit tests, linting, and database migration checks before executing rolling container replacements, guaranteeing uninterrupted attendance tracking during updates.

### DISASTER RECOVERY & REDUNDANCY

#### Automated Continuous Snapshots & WAL Archiving

PostgreSQL Write-Ahead Logging (WAL) and automated point-in-time recovery enable immediate database restoration with zero data loss. Encrypted multi-region cold backups run daily.

### SERVICE LEVEL AGREEMENT

#### 99.99% Availability & Priority Support

Enterprise contracts guarantee 99.99% cloud API uptime, 24/7 dedicated solutions engineering response within 15 minutes for critical incidents, and scheduled quarterly security audits.